

Session Objectives

- At the end of the session, the student should be able to:
 1. Describe the anatomical landmarks and the 3 anatomical triangles that can potentially cause thoracic outlet syndrome
 2. Deduce the most likely area of entrapment contributing to a patient's symptoms based on information gathered in the history
 3. Diagnose thoracic outlet syndrome based on patient history and physical exam findings
 4. Formulate an osteopathic manipulative treatment plan for a patient with thoracic outlet syndrome based on osteopathic structural considerations

Thoracic Outlet Syndrome (TOS)

- Group of diverse conditions with variable symptomology that results from compression of the neurovascular bundle
- What makes thoracic outlet syndrome (TOS) special?
 - A. Uncommon
 - B. Significant symptoms
 - C. Diagnostic challenge
 - D. Illustrates importance of anatomical knowledge and reasoning

Symptoms of TOS include the following:

- Pain in shoulder and/or upper extremity
 - Paresthesia (e.g. numbness, tingling, burning); Coldness
 - Pallor
 - Provoked by certain motions
 - Poor function (weakness, heaviness, fatigue, lack of endurance, atrophy)
 - Particular side
-
- Any general thoughts? (many sx's but somehow tied together, many conditions can cause this!)

Types of TOS

- **Neurogenic:** compression of the brachial plexus (C5-T1)
 - Most often lower plexus (C8 and T1)- severe weakness and muscle atrophy may result if chronic
- **Vascular**
 - **Venous:** compression of the subclavian vein or axillary vein
 - Arterial

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What is the *most common* type of TOS (Neurogenic) and its *percentage of cases* (>90%)?

History taking: being the detective



What should we look for on the history?

- A. Types and distribution of symptoms (including laterality)
- B. Provoking factors (movement induced typically)
- C. Timing of symptoms
- D. Daily activities; any change *before and since* symptom onset?
- E. Trauma

History taking: being the detective

What *else* should we look for on the history (or be mindful of)?

- A. Smoking status
- B. Advanced age
- C. Severe, rapidly progressive symptoms
- D. Non-musculoskeletal symptoms
- E. Constitutional symptoms and red flags (eg. weight loss, night sweats, dyspnea, hemoptysis)



Anatomical considerations in TOS: What's in a name?

Consider other names that have been used to refer to TOS:

- Cervical rib syndrome
- Scalenus anticus syndrome
- Costoclavicular space syndrome
- First rib syndrome
- Paget-von Schroetter syndrome (aka venous TOS, primary effort thrombosis, etc.)

Anatomical considerations in TOS:

Three primary regions of entrapment

- Interscalene triangle (aka scalene triangle)
 - Brachial plexus and Subclavian artery pass through
 - Most common site of **brachial plexus impingement**
- Costoclavicular space
 - Brachial plexus, Subclavian artery and Subclavian vein pass through
 - Most commonly compressed at this site is **subclavian vein**
- Subcoracoid space (aka retropectoralis space or subcoracoid pectoralis minor space)
 - Brachial plexus, Axillary artery and Axillary vein pass through
 - **Neurovascular bundle compression** almost as common as in interscalene triangle

Symptoms/signs and triggers for Neurogenic TOS

- **About 90% of total cases of TOS; *most often* from compression of the brachial plexus in the anterior scalene triangle**

- Upper extremity paresthesia (98%)
- Neck pain (88%)
- Trapezius pain (92%)
- Shoulder and/or arm pain (88%)
- Supraclavicular pain (76%)
- Chest pain (72%)
- Occipital headache (76%)
- Paresthesia in fingers (58% in all; 26% in 4th and 5th only)
- Muscle atrophy
- Scalene tenderness

- Repetitive activity (esp. if vigorous or overhead)
- Heavy lifting
- Prolonged computer work
- Driving

Symptoms/signs and triggers for Venous TOS (aka Paget-von Schroetter syndrome or “Effort Thrombosis”)

- **10-15% of cases; from compression of the subclavian vein within the costoclavicular space**

- Forearm fatigue within minutes of using arm may be present
- UE **swelling**, cyanosis, heaviness
- UE paresthesia and/or pain
- Swelling in shoulder, supraclavicular, or chest may be present; venous patterning if vein stenosis/occlusion
- Consider thrombosis

- Repetitive activity (esp. if vigorous or overhead)
- Heavy lifting
- Spontaneous vs trauma-induced
- Extrinsic compression (eg. tumor)

Symptoms/signs and triggers for Arterial TOS

- **2-5% of cases; from compression of the subclavian artery within the scalene triangle**

- Pallor
- Pain
- Paresthesia
- Coldness
- Decreased BP
- Diminished or absent pulse
- Sxs not elicited by provocative maneuvers
- Scalene not tender
- Consider aneurysm (eg. aneurismal pulse) or thrombus

- Almost always associated with a cervical rib or anomalous rib
- Not work or trauma related
- However, vigorous activity may be a factor in some (eg. athletes)

Based on the clinical historical and anatomical considerations of TOS, what should the immediate focus of your clinical evaluation be?

- A. Identify potentially serious or limb threatening conditions
- B. Neurological status
- C. Vascular status

Adson's test: What does it test?

- Purpose:

- “to determine the state of the subclavian artery, which may be compressed by an extra cervical rib or by tightened anterior and middle scalene muscles” (*FOM, 4th ed: Osteopathic considerations in Neurology*)
- For “brachial plexus impingement between (the) anterior and middle scalene” (*The 5-Minute Osteopathic Manipulative Medicine Consult, 2e > Specialized Tests: Head, Cervical, Lumbar, Lower Extremity, Innominate, Sacrum and Upper Extremity*)

Diagnostic methods for TOS

- Guided by your history and physical exam
- Methods include:
 - Radiographs: spine, chest, shoulder, upper extremity
 - MR, CT, MRN
 - Ultrasound
 - Electrophysiological studies (EMG/NCS)
 - Arteriography, angiography
 - Hemodynamic testing
 - Local anesthetic injection into anterior scalene
 - 94% positive outcome after surgery in those with positive response to block (vs. 50% response in those who failed block)

Diagnostic goals

- Determine if alternative diagnosis can better explain the patient's symptoms (most, if not all)
 - Think proximal and distal compressive neuropathies (Spurling maneuver, Tinel's sign, Phalen's sign) and other causes (eg. cervical arthritis) that may cause sx's
 - Think of somatic dysfunctions that can impede anatomical relationships and anatomical functioning
- Determine if an alternative diagnosis is *coincidental (concomitant)* or *causative* to the patient's symptoms/signs
- We can then treat the condition by addressing the cause(s), and improving symptoms and function

Other important structures to consider:

- Posture
- Cervical spine (C3-6 for anterior scalenes; C2-7 for middle scalenes)
- Clavicle
- Thoracic spine (sympathetic innerv. to UEs; impact on ribs)
- Upper extremity
 - Clavicle
 - Shoulder, Scapula
 - Elbow, wrist, hand
 - Soft tissue findings (eg. changes in color, temp, hair pattern, texture, atrophy, etc.)

Management:

- Conservative
 - Physical rehabilitation (posture, stretching, strengthening)
 - OMT
 - Activity modification, ergonomics
 - Stress reduction
 - Weight control
 - Pharmacological (eg. NSAIDs, muscle relaxants, antidepressants, anticonvulsants; *thrombolysis, heparin*)
 - Injection (analgesics, steroid, including botulinum toxin type A)
 - Patient education and monitoring are key
 - Adequate trial of 4-6 months (usually)

Management:

- Surgical (vascular surgeon typically)
 - Failed conservative management
 - Arterial or Venous TOS- often initial management is surgical
 - Neurogenic TOS- rib 1 resection (often with scalenectomy or scalenotomy)
 - Pros: “excellent” results in 95% of cases in nTOS; >95% patency rate in vTOS 5 years after surgery
 - Cons: complications can occur (eg. brachial plexus injury)

An osteopathic approach-continued

- Techniques that can be used include
 - Myofascial release (direct or indirect)
 - Soft tissue
 - Balanced ligamentous tension
 - Counterstrain
 - HVLA

In all cases, think through indications and contraindications, and adapt the technique to the patient

Summary

1. Thoracic outlet syndrome is an uncommon but important diagnosis to consider in patients with appropriate symptoms, especially if they are severe, worsen, or become chronic
2. It affects both the young (eg. teenagers) and old (50-60). In the young to middle aged, ask about vigorous or repetitive upper body activities, especially involving motions above shoulder level.
3. Diagnosis is challenging, but knowing the prevalence of the various types of TOS, and performing a detailed history and physical exam, including the osteopathic structural exam, can help you to determine the cause and aid in its management
4. Given the physical causes of thoracic outlet syndrome, the osteopathic manipulative approach of using look-feel-move can (and should) be incorporated into your evaluation to enhance your ability to diagnose and treat modifiable factors in the patient's care.